

# Current Landscape of the Real-World Asset (RWA) Tokenization Ecosystem

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## Abstract

The tokenization of real-world assets (RWAs) represents a pivotal development at the intersection of traditional financial markets and decentralized finance. This study undertakes a systematic examination of the RWA ecosystem through detailed due diligence of more than 180 products spanning six major asset classes: private credit, stocks, global bonds, commodities, institutional funds, and U.S. Treasuries. Our methodology combines a standardized review of legal structures and custodial arrangements with empirical analyses of on-chain transaction data. The analysis reveals that the RWA market, while still nascent, has already reached a scale exceeding \$25 billion and exhibits strong concentration, with dominant platforms accounting for over 80% of market share in several segments. Tokenization has facilitated fractional ownership and contributed to the establishment of novel legal-technical frameworks, particularly Swiss-based special purpose vehicles (SPVs) and U.S.-based statutory trusts. However, the oft-cited promise of enhanced on-chain liquidity remains largely unfulfilled. Instead, the data reveal a persistent divergence between the theoretical advantages of 24/7 trading and the empirical reality of limited secondary-market activity and predominantly passive holding patterns. By providing a comprehensive survey of the current RWA landscape, this research identifies the structural frictions that must be addressed for the market to evolve from an experimental proof-of-concept into a mature, institutionally integrated asset class.

**Keywords:** Blockchain, Real World Assets, Tokenization.

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# 1 Introduction

The tokenization of real-world assets (RWAs) represents a recent development at the intersection of traditional finance (tradFi) and decentralized finance (DeFi). At its core, this process converts the rights to an underlying asset into digital tokens recorded on a blockchain, thereby embedding traditionally off-chain instruments within programmable financial infrastructure. The proclaimed value proposition is compelling: to unlock the vast value of traditional assets, estimated in the hundreds of trillions, by enhancing liquidity, enabling fractional ownership, and providing 24/7 global market access through transparent and programmable infrastructure [Karayaneva \(2024\)](#). Under this core value proposition, the RWA market has attracted significant attention and witnessed tremendous growth. What began as a niche proof-of-concept has rapidly evolved into a multi-billion-dollar ecosystem: By mid-2025, the total market capitalization of tokenized RWAs—excluding stablecoins—had surpassed \$24 billion, attracting participation from both crypto-native investors and established institutional actors [Redstone Finance \(2025\)](#).

However, the monolithic term “RWA market” obscures a highly fragmented and heterogeneous ecosystem in reality. Indeed, different parts of the ecosystem has not been evolving uniformly; rather, the “RWA market” comprises distinct asset classes each developing at different speeds and under disparate structural and regulatory paradigms. For instance, the private credit sector has emerged as a functional and rapidly maturing market, leveraging established legal wrappers to deliver tangible yield to investors. In stark contrast, the tokenized stocks sector, despite its theoretical promise, remains largely a collection of illiquid, proof-of-concept assets with negligible secondary market activity. This divergence highlights a critical gap in the existing discourse: a comprehensive, cross-asset class analysis is required to move beyond anecdotal evidence and develop a unified understanding of the market’s actual state.

This study addresses this gap by providing a systematic, multi-asset class analysis of the RWA landscape. Our contribution is threefold. First, we present a comprehensive synthesis of over 180 individual products across private credit, equities, global bonds, commodities, institutional funds, and U.S. Treasuries, applying a standardized due diligence framework to enable direct comparison. Second, we identify and analyze the prevalent legal structures support RWAs, including the Swiss DLT Act-compliant Special Purpose Vehicles (SPVs) favored by European issuers and the Regulation S-based trusts utilized for non-U.S. investor access. Third, we elevate product-level observations to identify empirical patterns and potential risks, including a pervasive problem of custodial opacity, the paradox of on-chain assets with little on-chain liquidity, and the critical role of regulatory arbitrage in the market’s current architecture.

The main data sources for our analysis comes from [rwa.xyz](#) by the end of 2025 summer. Since the RWA space has been rapidly evolving, it is expected that many quantitative results will change over time. However, we believe that the angles we take for the analysis and the main themes identified in the paper are likely to have lasting impact, and the results we summarize by the end of 2025 summer would be a useful snapshot for future retrospective studies.

**Related literature.** The RWA space is relative new, and there are still few academic studies that directly look into RWA. One exception is [Malinova and Park \(2023\)](#) who examine the potential advantages and challenges of tokenizing assets on public permissionless blockchains. [Cong, Landsman, Rabetti, Zhang and Zhao \(2025\)](#) focus on the 24/7 continuous trading feature of tokenized stocks, and relate the token price movements to the underlying stocks.<sup>1</sup> Both studies, however, mainly focus on tokenized stocks, while our paper cover all asset classes including U.S. Treasuries, non-U.S. government bonds, corporate bonds, private

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<sup>1</sup>For earlier studies on after-hours stock trading, see e.g., [Barclay and Hendershott \(2003\)](#) and [Li \(2016\)](#).

credit (the largest asset class by the time of study), commodities, institutional funds, and stocks. One notable exception is stablecoin, which we do not include in our study, since it already features a large volume of related studies.<sup>2</sup>

A study on RWA also naturally relates to a large literature on tokenomics, which is the process of creating new assets on a blockchain instead of representing existing assets on-chain. Notable theoretical contributions include contemporaneous studies by [Cong, Li and Wang \(2021\)](#), [Bakos and Halaburda \(2022\)](#), [Sockin and Xiong \(2023\)](#), and [Li and Mann \(2025\)](#), where a coordination problem around multiple equilibria is highlighted. There are also many studies on various aspects of initial coin offerings (ICOs): [Goldstein, Gupta and Sverchkov \(2019\)](#) show that utility tokens can limit the rent-seeking activities of two-sided platforms with market power while preserving efficiency gains due to network effects; [Catalini and Gans \(2018\)](#) show that dynamic pricing can elicit consumers’ willingness to pay; [Chod and Lyandres \(2018\)](#) show that an ICO can facilitate risk-sharing without dilution of control; [Canidio \(2018\)](#) studies the tension between ex-ante financing and ex-post incentives; [Garratt and van Oordt \(2019\)](#) investigates how ICOs change entrepreneur incentives compared to debt and venture capital financing; [Malinova and Park \(2018\)](#) compare tokens with equities; and [Lee and Parlour \(2022\)](#) show that when competition jeopardizes upfront cost recoupment, ICOs (and crowdfunding in general) mitigate underinvestment through direct funding from consumers. Empirically, [Howell, Niessner and Yermack \(forthcoming\)](#) document many important features of the ICO structure. [Kostovetsky and Benedetti \(2018\)](#) document high ICO returns. [Lee, Li and Shin \(2018\)](#) confirm wisdom of the crowd in the ICO

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<sup>2</sup>For a non-exhaustive list of recent works on stablecoins, see e.g., [Barthelemy, Gardin and Nguyen \(2018\)](#); [Griffin and Shams \(2020\)](#); [Zhao, Li and Yuan \(2021\)](#); [Kozhan and Viswanath-Natraj \(2021\)](#); [Li and Mayer \(2021\)](#); [Liao and Caramichael \(2021\)](#); [Lyons and Viswanath-Natraj \(2021\)](#); [Adams and Ibert \(2022\)](#); [d’Avernas, Maurin and Vandeweyer \(2022\)](#); [Kim \(2022\)](#); [Routledge and Zetlin-Jones \(2022\)](#); [Uhlig \(2022\)](#); [Aldasoro, Ahmed and Duley \(2023\)](#); [Anadu, Azar, Huang, Cipriani, Eisenbach, La Spada, Landoni, Macchiavelli, Malfroy-Camine and Wang \(2023\)](#); [Baughman and Flemming \(2023\)](#); [Bertsch \(2023\)](#); [Eichengreen, Nguyen and Viswanath-Natraj \(2023\)](#); [Flannery \(2023\)](#); [Gorton, Klee, Ross, Ross and Vardoulakis \(2023\)](#); [Liu, Makarov and Schoar \(2023\)](#); [Ma, Zeng and Zhang \(2025\)](#).

setting. [Davydiuk, Gupta and Rosen \(2023\)](#) investigate the extent to which token retention signals quality. [Li and Yi \(2019\)](#) evaluate smart beta strategies in crypto asset investment. [Adhami, Giudici and Martinazzi \(2018\)](#) and [Amsden and Schweizer \(2018\)](#) look for ICO success determinants. [Deng, Lee and Zhong \(2018\)](#) analyze the effect of disclosure quality, governance mechanism, and team networks on ICO project progress. [Momtaz \(2018\)](#) offers a description of the ICO market evolution. [Hu, Parlour and Rajan \(2019\)](#) provide investment characteristics of 64 ICOs. See [Li and Mann \(2021\)](#) for a survey of the ICO literature.

The remainder of this paper is structured as follows. Section 2 provides a market-wide landscape analysis. Sections 3 through 8 dives into each of the six asset classes: Private Credit, Stocks, Global Bonds, Commodities, Institutional Funds, and US Treasuries. Finally, Section 9 synthesizes our findings, presents a forward-looking SWOT analysis, and outlines the critical questions that will define the future of real-world asset tokenization.

## 2 Overall Market Landscape

This section aggregates quantitative data from all analyzed products to provide a market-level overview, focusing on the scale, primary actors, and technical foundations of the current RWA ecosystem.

### 2.1 Market Size & Growth across Sectors

As of the third quarter of 2025, the tokenized real-world asset market has reached a total capitalization of approximately \$24 billion, exhibiting a consistent growth rate of approximately 6.3% month-over-month [Redstone Finance \(2025\)](#). This aggregate figure, however, masks a significant heterogeneity across asset classes. Indeed, the market is not a monolith but a collection of distinct ecosystems, each with its own level of maturity and scale.

The dominant asset class by value is Private Credit, which constitutes approximately

58% of the market’s on-chain value. This is followed by U.S. Treasuries and Global Bonds that have experienced explosive growth, surging more than 80% in 2025 to a market capitalization of \$7.4 billion, as crypto-native investors look for yield-bearing alternatives to stablecoins. The Commodities sector, primarily composed of tokenized gold, represents a mature but more stable segment with a market capitalization of approximately \$1.74 billion. In contrast, Tokenized Stocks and Institutional Funds, despite representing a high volume of individual products, collectively account for less than 6% of the total market capitalization (approximately \$1.26 billion combined), underscoring the performance gap between established, institutional-grade offerings and speculative proof-of-concept tokens.

## **2.2 Issuer and Platform Dominance**

The real-world asset market is currently largely dominated by several asset-class-specific oligopolies. A small number of issuer-platforms control the vast majority of on-chain value and product offerings, creating significant market concentration. This structure is not uniform across the ecosystem; rather, each major asset class exhibits its own distinct set of dominant actors, as to be detailed in subsequent sections.

This concentration is most pronounced in the Commodities and Global Bonds sectors. The tokenized commodities market, valued at approximately \$1.74 billion, functions as a near-duopoly, with two issuers, Paxos (PAXG) and Tether (XAUT), collectively accounting for over 98% of the total market capitalization. A similar structure is observed in the global bonds market, where a single issuer, Spiko, controls approximately 84% of the sector’s on-chain value. In the Institutional Funds space, dominance is manifested at the infrastructure level, where the platform Securitize underpins a significant portion of the total market value.

The Tokenized Stocks sector presents a more nuanced picture of concentration, with dominance split between market value and product count. A single issuance, Exodus Movement, Inc. (EXOD), accounts for over 60% of the sector’s total market capitalization. Platforms like

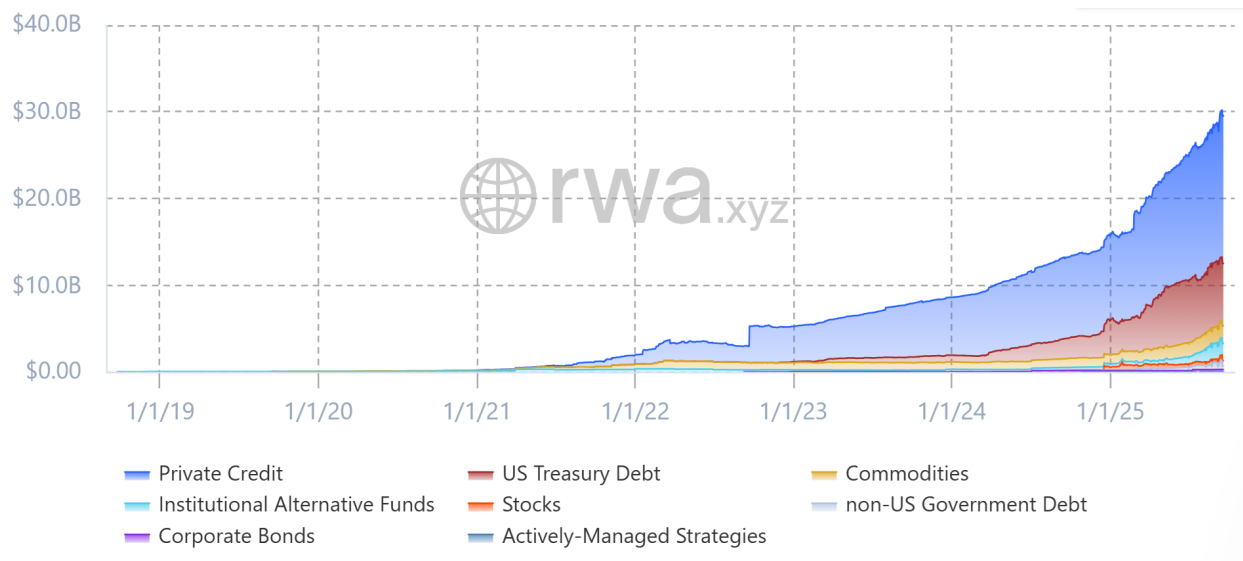


Figure 1: A time-series chart illustrating the growth of the total RWA market capitalization (excluding stablecoins) over the past 6 years with contribution of each major asset class. Source: [app.rwa.xyz/](http://app.rwa.xyz/).

Dinari and Backed Finance are dominant by product count, despite controlling a relatively small fraction of total asset value. This bifurcation indicates two distinct market strategies: a high-value, single-asset approach versus a broad, retail-focused “long-tail” strategy.<sup>3</sup>

The primary implication of this market-wide concentration is the introduction of significant systemic risk. The operational, legal, or technical failure of a single dominant issuer or platform could have a disproportionate and cascading impact on its respective asset class. These high barriers to entry, driven by regulatory compliance costs and the necessity of establishing institutional trust and custodial relationships, favor established entities and may inhibit the emergence of a more decentralized and resilient issuer landscape.

<sup>3</sup>See product offering details at [Backed Finance \(2025\)](#); [Dinari, Inc. \(2025a,c,b\)](#); [Exodus Movement, Inc. \(2025\)](#).

## 2.3 Blockchain Ecosystem Analysis

The technical foundation of the real-world asset market is increasingly multichain, though a clear hierarchy of platforms has emerged based on asset value, product count, and intended use case. The choice of a specific blockchain is not arbitrary but a strategic decision reflecting an issuer’s priorities, ranging from institutional-grade security and liquidity to retail-focused cost efficiency.

Ethereum remains the predominant blockchain by total value locked (TVL), serving as the primary settlement layer for a significant majority of high-value RWA products, particularly in the private credit and global bond sectors. This preference is driven by its deep liquidity, extensive DeFi infrastructure, robust security model, and established reputation among institutional actors. For issuers targeting institutional capital, Ethereum’s battle-tested environment provides a degree of trust and composability that currently remains unmatched.

Concurrently, a strategic migration to alternative Layer 1 (L1) and Layer 2 (L2) networks is evident, particularly for products prioritizing high throughput and low transaction costs. High-performance L1s such as Solana and Algorand have been selected for specific, large-scale issuances where transaction speed is critical. The proliferation of products on L2 networks, most notably Arbitrum, Base, and Polygon, is a direct consequence of their low transaction costs. Issuers pursuing a “long-tail” strategy, such as Dinari with its extensive catalog of tokenized stocks, heavily favor these platforms to make the minting and trading of low-capitalization assets economically viable for a retail audience.

This fragmentation of the RWA ecosystem across multiple blockchains, on the one hand, demonstrates market maturity, as issuers select the optimal technical environment for their specific product. On the other hand, it introduces significant challenges regarding interoperability and fragmented liquidity. For the market to function as a cohesive whole, robust and secure cross-chain infrastructure will be essential to bridge these increasingly siloed

ecosystems.<sup>4</sup>

## 2.4 Prevailing Structures and Regulatory Frameworks

The legal and structural underpinnings of the real-world asset market are as varied as its technical implementations, with issuers strategically selecting jurisdictions and legal wrappers to balance regulatory compliance and operational flexibility. This process of “regulatory arbitrage” has led to the emergence of several dominant structural models, each with distinct implications for investor protection and systemic risk.

A primary model, particularly prevalent in the Tokenized Stocks and Global Bonds sectors, uses Special Purpose Vehicles (SPVs) domiciled in jurisdictions with progressive digital asset legislation. Switzerland, with its DLT Act, has become a favored jurisdiction, enabling issuers to create legally recognized entities to hold the underlying assets. This framework provides a clear legal basis for the tokens representing ownership claims. Similarly, issuers leverage established fund structures in jurisdictions like Luxembourg and France, often under the UCITS or MiFID II regimes, to offer products with robust, recognized investor protections [European Union \(2009, 2014\)](#).

A second model, commonly employed by issuers targeting non-U.S. investors, uses U.S.-based statutory trusts combined with exemptions under the Securities and Exchange Commission’s (SEC) Regulation S. This structure allows issuers to offer financial exposure to U.S.-domiciled assets while legally barring U.S. persons from participation, thereby avoiding the stringent registration requirements of the SEC. While effective for global distribution, this model often results in weaker legal claims for token holders compared to direct equity

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<sup>4</sup>For the large volume of studies on blockchain interoperability, see for example [Hardjono, Lipton and Pentland \(2018\)](#); [McCorry, Buckland, Yee and Song \(2021\)](#); [Zamyatin, Al-Bassam, Zindros, Kokoris-Kogias, Moreno-Sanchez, Kiayias and Knottenbelt \(2021\)](#); [Neulinger \(2022\)](#); [Pillai, Biswas, Hóu and Muthukumarasamy \(2022\)](#); [Thyagarajan, Malavolta and Moreno-Sanchez \(2022\)](#); [Bugnet and Zamyatin \(2022\)](#); [Scaffino, Aumayr, Avarikioti and Maffei \(2023\)](#); [Lee, Murashkin, Derka and Gorzny \(2023\)](#); [Scaffino, Aumayr, Avarikioti and Maffei \(2023\)](#); [Yan, Lu, Agrawal, Li, Diao and Zhang \(2025\)](#); [Li and Wu \(2025\)](#).

ownership [U.S. Securities and Exchange Commission \(n.d.\)](#)

A third, more nascent approach involves the use of Decentralized Autonomous Organizations (DAOs) paired with legal wrappers in offshore jurisdictions such as the Marshall Islands or the Cayman Islands. These structures offer maximum operational flexibility and align with the ethos of decentralized governance but typically provide the least amount of formal, state-enforced investor protection.

A common thread across these models is issuers' strategic calculations to balance speed-to-market and regulatory efficiency. However, the practice has also led to a fragmented and complex legal landscape where the rights and recourse available to token holders depend heavily on the specific legal structure and jurisdiction of the issuing entity. This legal heterogeneity remains a significant barrier to the creation of a truly global, fungible, and trusted RWA market.

### **3 Asset Class Analysis: Private Credit**

This section compiles all of the data gathered regarding private credit from [rwa.xyz](#). Private credit is a type of loan, issued typically via cryptocurrencies, which can be beneficial for the provider or the acceptor based on the volatility of the cryptocurrency; that is to say, lenders will provide capital if they project that the cryptocurrency will increase in value, and borrowers will accept if they project that the cryptocurrency will decrease in value.

#### **3.1 Market Composition & Dominant Protocols**

The tokenized private credit market has evolved into a diverse and rapidly expanding segment of decentralized finance (DeFi), blending the yield-generation capabilities of traditional credit markets with the transparency and immutability of blockchain technology. At its core, the market is composed of multiple blockchain-based lending pools, each designed with vary-

ing degrees of decentralization, investor eligibility requirements, collateralization standards, and legal enforceability. Dominant protocols in this space include Maple Finance, Goldfinch, Centrifuge, PACT, Fortunafi, and Maven 11, each serving different investor segments and asset classes. Maple Finance is a clear leader, hosting multiple high-profile pools such as Syrup USDC, Syrup USDT, Maple Direct USDC1, High Yield Secured Lending USDC1, and the Maven 11 wETH Pool, with most of its offerings incorporated in Delaware and targeted at accredited or institutional investors. Goldfinch operates under a more permissionless model, offering global participation through its Senior Pool, which enables diversified lending to off-chain partners without requiring direct negotiation with borrowers. Centrifuge specializes in tokenized, asset-backed lending such as New Silver 2, Harbor Trade 2, and ConsolFreight 4. These products are typically backed by real-world collaterals like real estate debt or trade finance receivables and use tranching to cater to different investor risk appetites. PACT, operating on the Aptos blockchain, has demonstrated institutional-grade private credit scaling with rapid TVL growth, exemplified by its Berkeley Square pool. Fortunafi combines regulated private credit exposure with the benefits of tokenization, while Maven 11 offers both stablecoin-denominated and wETH-denominated credit pools, targeting institutional investors comfortable with either fiat-pegged or crypto-native yield exposure. We should note that our data source [rwa.xyz](https://rwa.xyz) does not cover relatively small RWA platforms, such as the real estate RWA platform RealT documented in [Mafrur \(2025\)](#).

### **3.2 Prevailing Structures and Regulatory Frameworks**

The prevailing legal structures across these products reveal a strong preference for Delaware entities, including corporations, LLCs, and trusts. Delaware’s favorable legal precedents, tax efficiency, and established frameworks for securities exemptions—most notably Regulation D for U.S. accredited investors and Regulation S for non-U.S. investors—make it the jurisdiction of choice. Special Purpose Vehicles (SPVs) are frequently employed in trade

finance-focused pools, including Harbor Trade 2, ConsolFreight 4, and Maven 11 USDC, to isolate assets and liabilities. On the other end of the spectrum, permissionless protocols like Goldfinch rely more on decentralized governance than on a central legal entity, leaving enforcement primarily to the borrower’s jurisdiction. Regulatory compliance remains central to most institutional-grade offerings, with many U.S.-based pools also leveraging Investment Company Act exemptions such as 3(c)(1), alongside strict KYC/AML enforcement and whitelisted wallet access. Investor access models vary widely, ranging from accredited-only pools with fully documented offering memorandums to permissionless products that prioritize accessibility, albeit with more uncertain legal recourse.

### **3.3 Economic Rationale & On-Chain Activity**

The economic rationale behind tokenized private credit is rooted in yield enhancement, operational efficiency, and market expansion. Typical APYs in this sector range between 7% and 10%, appealing to both crypto-native and traditional investors seeking higher returns than those offered by conventional fixed-income securities. Tokenization allows for significant capital efficiency by reducing operational overhead, enabling fractional ownership, and providing 24/7 global access to previously illiquid markets. Some protocols, like Centrifuge, offer tranching systems such as DROP and TIN to tailor risk exposure. Platforms such as PACT and Goldfinch are focused on democratizing access to private credit by lowering barriers to entry, whereas Maple and Fortunafi prioritize a compliant, institution-first approach. On-chain behavioral patterns reflect the sector’s dynamism: high TVL growth is common, with Berkeley Square’s PACT pool growing 57% in a single month and Fortunafi 1 expanding from \$150 million to \$250 million in Q1 2025. Many pools exhibit concentrated investor bases dominated by institutional funds, DAOs, and pool delegates, while liquidity cycles are closely tied to loan origination and repayment schedules. Transparency is a defining advantage, with most protocols publishing verified on-chain collateral addresses and loan

performance metrics.

### **3.4 Synthesis: Strengths, Risks, and Outlook for Private Credit**

Despite the promising growth trajectory, the sector’s strengths are counterbalanced by notable risks. On the positive side, tokenized private credit benefits from institutional-grade infrastructure, including professional custody services from Anchorage, BitGo, and Zodia; rigorous third-party audits from firms such as ThreeSigma and 0xMacro; and strong legal enforceability in jurisdictions like Delaware. Global reach is another defining characteristic, as tokenized structures allow fractionalized participation from investors worldwide, regardless of geography. On-chain transparency further boosts investor confidence by enabling real-time NAV tracking and detailed credit performance visibility. Product diversity is also a strength, encompassing stablecoin-denominated pools, wETH-based credit, U.S. real estate-backed lending, and trade finance-backed offerings.

However, risks remain substantial. Credit defaults, as seen in the Maven 11 wETH Pool, have directly impacted LP returns and underscore the necessity of robust borrower vetting. Regulatory uncertainty is a looming factor, especially for permissionless pools operating without explicit jurisdictional anchoring. Liquidity constraints are also common—many products restrict redemptions to fixed windows or epochs, creating potential bottlenecks during market stress. Stablecoin-backed pools remain vulnerable to peg instability, with USDC and USDT depegging events representing systemic risks. Furthermore, heavy concentration of capital among a few large investors can weaken resilience against sudden withdrawals or shifts in sentiment.

On a speculative note, we expect the outlook for tokenized private credit to remain strong in both the short and medium term. In the near future, growth is likely to be led by institutional compliant pools that pair high transparency with robust legal frameworks, ensuring both investor protection and regulatory acceptance. Medium-term developments

may include greater integration of off-chain legal enforcement tools, standardized borrower risk assessment frameworks, and the introduction of loss insurance products to mitigate default risk. In the longer term, the potential exists for a convergence between DeFi-native liquidity mechanisms and traditional securitization markets, creating a unified, on-chain private credit marketplace that blends the best of both worlds. Should this vision materialize, tokenized private credit could become a cornerstone of both digital and traditional capital markets, offering unparalleled efficiency, transparency, and accessibility while retaining the risk management rigor of traditional finance.

## **4 Asset Class Analysis: Stocks**

This section provides a detailed analysis of the Tokenized Stocks. It synthesizes findings from over 100 individual product due diligence reports to evaluate the market’s composition, its prevailing structural and economic models, and its current state of on-chain maturity.

### **4.1 Market Composition & Issuer Dominance**

The tokenized stock market, with a total capitalization of approximately \$489 million across more than 160 products, presents a landscape of extreme concentration and strategic divergence. The market is not a fragmented field of competitors but is instead an unbalanced oligopoly controlled by a handful of issuer-platforms, each pursuing a distinct model of tokenization.

A stark bifurcation exists between market leadership in terms of value and the number of product offerings. Dominance by market capitalization is absolute: a single issuance, Exodus Movement, Inc. (EXOD), accounts for over 60% of the asset class’s total value at approximately \$302 million. This single, regulated token effectively dictates the entire asset class’s perceived scale. In stark contrast, Dinari is the most prolific issuer by product count,

with a catalog of over 340 individual dShares. However, this extensive portfolio collectively represents less than 1% of the market’s total capitalization, indicating a “longtail” strategy that has yet to achieve significant capital formation. Issuers such as Backed Finance and Swarm occupy a middle ground, offering curated selections of several dozen blue-chip equities that, while more numerous than Exodus, still command a smaller share of the total market value.

The primary implication of this structure is a market that is simultaneously stable yet vulnerable at the top, and fragmented and inefficient at the bottom. The concentration of value in a single asset creates a systemic dependency, while the proliferation of micro-capitalization tokens with negligible liquidity inhibits the development of a cohesive and fungible secondary market. This dynamic suggests that while the technical and legal frameworks for tokenizing a wide array of equities exist, the market has not yet found a scalable model for attracting and distributing capital beyond a few flagship assets.

## **4.2 Prevailing Structures and Regulatory Frameworks**

The legal architecture of the tokenized equity market has largely consolidated into two jurisdiction-specific models, reflecting issuers’ reliance on regulatory arbitrage to reconcile compliance requirements with operational flexibility. These frameworks critically shape both the legal character of the token holder’s claim and the scope of investor protections provided.

The first and most prominent model, particularly for European-facing issuers, is the use of Special Purpose Vehicles (SPVs) domiciled in jurisdictions with progressive digital asset legislation. Switzerland, with its Federal Act on the Adaptation of Federal Law to Developments in Distributed Ledger Technology (DLT Act), has emerged as the favored jurisdiction for issuers like Backed Finance. This framework allows for the creation of legally distinct entities that hold the underlying equity, providing a clear legal basis for the tokens as tradable securities [Switzerland \(2021\)](#). A similar approach is taken in Liechtenstein under

its Blockchain Act (TVTG), which underpins the offerings from platforms like Swarm [Swarm Markets GmbH \(2025\)](#). This European model is often paired with prospectus-based offerings compliant with regional regulations, providing a robust compliance wrapper for non-U.S. investors.

The second dominant model involves the use of U.S.-based statutory trusts, primarily utilized by issuers like Dinari to target a global, non-U.S. audience. This structure is strategically combined with exemptions under the U.S. Securities and Exchange Commission’s Regulation S, which permits the offering of securities to non-U.S. persons without requiring full SEC registration (U.S. Securities and Exchange Commission, n.d.-b). In this framework, the token does not grant direct legal ownership of the underlying stock but rather a beneficial interest or contractual claim on the shares held by the trust. While legally effective for achieving global distribution, this model often results in weaker direct legal recourse for token holders and typically strips them of fundamental shareholder rights, such as voting.

A critical and pervasive flaw that transcends both models is the opacity of custodial arrangements. A systematic review of the product documentation reveals that the majority of issuers, including major platforms, fail to publicly disclose the specific third-party custodians responsible for holding the underlying shares. This lack of transparency represents a significant systemic risk, forcing investors to place their trust in the issuer’s claims rather than in verifiable, independent proof-of-reserves. This practice fundamentally undermines the core promise of blockchain-based transparency and exposes investors to unquantifiable counterparty risk.

### **4.3 Economic Rationale & On-Chain Activity**

The stated economic rationale for tokenizing equities centers on addressing three fundamental inefficiencies of traditional markets: democratizing access through fractionalization, enabling 24/7 global trading, and unlocking novel utility through decentralized finance (DeFi) com-

posability. The target investor profile, as evidenced by the widespread use of Regulation S exemptions and the strategic selection of popular U.S. technology and “meme” stocks, is overwhelmingly a non-U.S., crypto-native retail audience for whom direct brokerage access is often cumbersome or unavailable.

However, an empirical analysis of on-chain data reveals a profound disconnect between this stated value proposition and the market’s current reality. The asset class is characterized by an extremely passive, “buy-and-hold” transaction profile, with the vast majority of the 106 products analyzed showing zero or single-digit monthly transfer counts. The dominant on-chain events are not peer-to-peer trades but are almost exclusively mints and burns corresponding to primary issuance and redemption directly with the issuer’s centralized platform. Peer-to-peer transfers, the definitive signifier of a functioning secondary market, are exceedingly rare.

This on-chain data tells an unambiguous story of a market failing to achieve its core economic purpose. The promise of 24/7 liquidity remains almost entirely theoretical, creating a “liquidity paradox” where on-chain assets possess no on-chain liquidity. This analysis confirms that tokenized stocks are not currently functioning as dynamic, liquid financial instruments. Instead, they operate as static, illiquid digital certificates of ownership, accessible only through platform-gated portals, thereby failing to deliver on their most critical promised innovation.

#### **4.4 Synthesis: Strengths, Risks, and Outlook for Tokenized Stocks**

The tokenized stocks asset class, in its current state, is best understood as a successful proof-of-concept for legal and technical issuance, but a functional failure in market creation. Its primary strengths lie in the establishment of replicable, jurisdiction-aware legal blueprints, namely the Swiss SPV and the U.S. Trust/Regulation S models, that have effectively solved the challenge of compliant, cross-border issuance. In doing so, these platforms have suc-

cessfully democratized access to global equities for a non-U.S. retail audience, providing economic exposure where traditional brokerage channels are often inaccessible.

However, these achievements are overshadowed by profound systemic risks and limitations that call into question the viability of the current market structure. The most critical failure is the liquidity paradox: despite being built on infrastructure designed for 24/7 trading, the vast majority of these assets exhibit zero on-chain secondary market activity. This renders the core value proposition of enhanced liquidity entirely theoretical and relegates these tokens to the status of illiquid, platform-gated digital certificates. Compounding this is a pervasive custodial opacity, where the failure of major issuers to disclose custodians or provide independent, verifiable proof-of-reserves creates a market built on unverifiable trust and exposes investors to significant counterparty risk. Finally, the market is defined by a high degree of platform centralization, where a technical, legal, or operational failure at one of a handful of dominant issuers could have a cascading impact across a significant portion of the ecosystem.

On a speculative note, the future trajectory of the tokenized stocks market seems to be at a critical inflection point. Its evolution from a collection of novel but illiquid pilot projects into a mature asset class is contingent on solving these fundamental challenges. Future growth may not only depend on the creation of more tokenized assets, but also on the development of robust secondary market infrastructure, such as incentivized automated market makers (AMMs) or partnerships with regulated exchanges. Furthermore, the market may not be able to attract significant institutional capital until issuers commit to more transparency regarding custodial arrangements or provide independent, on-chain verifiable audits. Without these foundational shifts, the tokenized stock market risks remaining a niche experiment that, while legally innovative, failed to deliver on its economic promise.

## 5 Asset Class Analysis: Global Bonds

The tokenized global bond market remains highly concentrated among a few issuers. Spiko leads with over \$220 million in its flagship EU T-Bills Money Market Fund, representing the majority of market capitalization across all products. Other notable issuers include Backed Finance, OpenTrade, and Etherfuse. France and Switzerland dominate regulated issuance. Ethereum, Solana, and Avalanche carry the bulk of on-chain activity. While some funds target a broad institutional base, others restrict participation to non-U.S. investors or accredited profiles, limiting retail exposure.

### 5.1 Market Composition & Issuer Dominance

Tokenized global bonds represent a rapidly evolving asset class that seeks to modernize the fixed-income market by replacing legacy settlement and distribution infrastructure with blockchain-based systems. As of mid-2025, the market is still in an experimental stage, but activity is expanding. According to the Bank for International Settlements, more than sixty tokenized bond issuances have occurred to date, representing roughly USD 8 billion in value across sovereign, supranational, and corporate issuers. The market is not evenly distributed, with Securitize, Inc. as a clear market leader. Other major institutions such as Societe Générale, Goldman Sachs, HSBC, UBS, and BNP Paribas have also launched tokenized bond products, while Switzerland’s SIX Digital Exchange has pioneered fully regulated digital bond listings, including UBS’s CHF 375 million digital bond. The dataset analyzed for this report covers fifteen tokenized global bond products issued across key jurisdictions such as France, Switzerland, Luxembourg, Mexico, the United States, and the Cayman Islands, with median market capitalizations below USD 6 million. Although relatively small, these offerings demonstrate meaningful diversity in issuer type, regulatory structure, and blockchain network usage.

## 5.2 Prevailing Structures and Regulatory Frameworks

Legal and structural models for tokenized bonds vary widely, reflecting the fragmented regulatory landscape. In the European Union, frameworks such as the UCITS Directive, the MMF Regulation, MiFID II, and the DLT Pilot Regime provide explicit legal pathways for tokenized investment products [European Union \(2009, 2014\)](#); [European Securities and Markets Authority \(2023\)](#). Switzerland has advanced the sector with its Digital Ledger Technology Act, which creates enforceable legal recognition for tokenized securities [Switzerland \(2021\)](#). A taxonomy of tokenization distinguishes between complete tokenization, where tokens confer direct and legally binding rights; indirect tokenization, where tokens reference off-chain claims; and so-called “digital twins,” which represent instruments without direct legal standing. To provide market guidance, the International Capital Market Association (ICMA) has issued a comprehensive DLT Bonds Reference Guide [International Capital Market Association \(2024\)](#). In the dataset analyzed here, three products adopt regulated EU structures such as UCITS, while nine rely on crypto-specific regimes like Switzerland’s DLT Act or Luxembourg’s MiFID II, compliant frameworks. This blend illustrates how issuers are experimenting with both traditional compliance regimes and specialized crypto-asset regulations to establish legitimacy.

## 5.3 Economic Rationale & On-Chain Activity

The economic rationale for tokenizing bonds rests on efficiency, accessibility, and programmability. Blockchain settlement reduces time-to-finality from days to near-instant; for example, Hong Kong’s first digital government bond reduced settlement to T+1. Empirical data from the BIS suggests tokenized bonds may feature lower bid-ask spreads, faster issuance processes, and broadly comparable yields to conventional debt securities. By enabling fractional ownership, tokenization allows smaller investors to participate in traditionally inac-

cessible fixed-income markets. Programmability, through smart contracts, facilitates automated coupon payments, collateral management, and secondary transfers, thereby reducing operational friction. Recent momentum in the sector is notable: tokenized Treasury and money-market products grew by more than 80 percent in 2025 to USD 7.4 billion, driven largely by demand for faster settlement cycles and the ability to use tokenized debt as collateral in on-chain finance. However, the products in this dataset remain modest in size, with limited secondary trading and inconsistent disclosure regarding redemption mechanics or peg maintenance. Still, many feature clear custodial relationships, transparent fee structures, and stable issuance practices, reflecting a maturing operational baseline.

## **5.4 Synthesis: Strengths, Risks, and Outlook for Global Bonds**

Overall, tokenized global bonds offer clear strengths alongside meaningful risks. Their strengths include efficiency gains in settlement and transaction costs, enhanced transparency through blockchain-based auditability, expanded access for investors via fractionalization and low entry thresholds, and compliance options under both traditional and crypto-specific regulatory regimes. Risks include regulatory uncertainty, especially in jurisdictions without clear frameworks; low adoption and illiquid secondary markets that limit exit options for investors; and unresolved technical challenges around interoperability, custodial trust, and oracle reliability. Moreover, transparency around redemption and peg mechanisms remains uneven, creating uncertainty for holders. Despite these limitations, the outlook for tokenized global bonds is constructive. Institutional uptake is growing, with major asset managers and regulated exchanges piloting products. Regulatory convergence through frameworks like MiCA, the DLT Pilot Regime, and ICMA guidance is likely to harmonize standards and reduce uncertainty. Infrastructure developments such as the Canton Network and Switzerland's SDX suggest that liquidity and cross-border usability may improve substantially in the coming years. Ultimately, while tokenized bonds remain in an early experimental phase, their

trajectory points toward a more efficient, transparent, and globally accessible fixed-income market.

## **6 Asset Class Analysis: Commodities**

Commodities represent one of the oldest and most fundamental asset classes in global finance. In the context of tokenization, commodities, particularly gold, have emerged as a leading category of real-world assets (RWAs) brought onto blockchains. The rationale is straightforward: commodities are tangible, globally traded, and well understood by investors, making them natural candidates for tokenized representations. As of mid-2025, gold dominates the tokenized commodity landscape, with products like Paxos Gold (PAXG) and Tether Gold (XAUT) leading the sector. This asset class sits at the intersection of traditional safe-haven investing and blockchain innovation, promising accessibility, programmability, and efficiency, while still contending with challenges of liquidity, regulation, and trust in custodians.

### **6.1 Market Composition & Issuer Dominance**

Tokenized commodities have grown into a notable segment within the broader real-world asset (RWA) ecosystem, reaching an estimated market size of about \$45 billion in 2024. Despite this expansion, they remain far smaller than tokenized treasuries and fiat-backed stablecoins, which dominate the RWA sector. Within commodities, gold is by far the leading underlying asset, and the market is highly concentrated among a small number of issuers. Paxos Gold (PAXG), issued by Paxos Trust Company, is the largest product, with a market capitalization of approximately \$856 million, followed by Tether Gold (XAUT) at \$679 million. Together, these two tokens represent more than 75 percent of the entire tokenized gold market. Other issuers such as Comtech Gold (CGO), VNX Gold (VNXAU), WisdomTree Gold (WTGOLD), and the Aurus suite of tokens for gold, silver, and platinum occupy much

smaller positions. This concentration underscores the dominance of Paxos and Tether in shaping both liquidity and market standards within the commodity token space.

## **6.2 Prevailing Structures and Regulatory Frameworks**

Tokenized commodities rely on diverse legal, regulatory, and structural frameworks, which vary significantly across issuers. Paxos Gold is issued by Paxos Trust Company in the United States and regulated under the New York Department of Financial Services (NYDFS). It is notable for its monthly attestations and regulated custodian arrangements. Tether Gold, in contrast, is domiciled in the British Virgin Islands and registered as a Money Services Business with the U.S. Financial Crimes Enforcement Network (FinCEN). Other issuers follow distinct models: Comtech Gold is based in the UAE’s Dubai Airport Free Zone; Aurus Technologies Ltd issues its tokens under a DMCC license in the United Kingdom and Dubai; WisdomTree Digital Holdings operates under NYDFS supervision in the United States; and VNX Commodities AG functions under the Liechtenstein Blockchain Act. Custody practices typically involve established institutions, though disclosure quality varies. These structural and legal variations highlight the fragmented regulatory environment in which commodity tokens operate.

## **6.3 Economic Rationale & On-Chain Activity**

The primary economic rationale behind tokenized commodities lies in their ability to provide fractional and accessible exposure to physical commodities, especially gold, without the logistical burdens of direct ownership. Investors can trade small units, use them as collateral, or integrate them into decentralized finance (DeFi) applications. Tokenized commodities also function as digital-native inflation hedges and diversification tools. Despite this potential, on-chain activity remains highly uneven. Paxos Gold and Tether Gold demonstrate mean-

ingful adoption, with monthly transfer volumes of \$896 million and \$549 million respectively. By contrast, smaller issuers such as VNX, WisdomTree, and Aurus tokens record minimal monthly activity. These disparities reveal a structural liquidity challenge: participation is often limited, secondary trading is shallow, and regulatory whitelisting requirements restrict market breadth. While tokenization introduces transparency and programmability, liquidity barriers remain a critical obstacle to their broader economic utility.

## **6.4 Synthesis: Strengths, Risks, and Outlook for Commodities**

The tokenization of commodities offers several clear strengths. By lowering entry barriers, these products enable fractional participation in markets historically limited to large investors. They integrate seamlessly into blockchain-based systems, allowing for collateralization and composability in DeFi, while leading issuers such as Paxos set a high bar for transparency and regulatory compliance. However, tokenized commodities also face significant risks. Liquidity is concentrated in a few tokens, leaving smaller projects stagnant. Regulatory fragmentation creates uncertainty, as issuers operate under very different legal frameworks. Counterparty exposure to issuers, custodians, and auditors remains a critical consideration. On a speculative note, the outlook for tokenized commodities is cautiously optimistic. Market growth is likely as regulatory clarity improves. Innovations such as tiered token architectures and hybrid marketplaces may address current liquidity constraints, while increased institutional participation could expand depth and adoption. Overall, tokenized commodities are positioned to play a growing role within the RWA ecosystem as infrastructure, regulation, and liquidity mechanisms mature.

## 7 Asset Class Analysis: Institutional Funds

Institutional funds represent one of the most advanced use cases for tokenized real-world assets (RWAs). These products bridge the gap between traditional finance (TradFi) structures and decentralized finance (DeFi) infrastructure by offering exposure to asset classes such as private credit, venture capital, and private equity through blockchain-based tokens. As of June 20, 2025, the tokenized institutional fund market demonstrates strong growth momentum, with established players like Superstate, Securitize, and Republic introducing institutional-grade products that comply with securities regulations while leveraging the efficiencies of public blockchains.

### 7.1 Market Composition & Platform Dominance

The tokenized institutional fund market is currently dominated by a handful of large funds. The Superstate Crypto Carry Fund (USCC) leads with approximately \$157.8 million under management, followed by Blockchain Capital’s BCAP fund at \$148.4 million and the Apollo Diversified Credit Fund (ACRED) at over \$97.2 million. A defining feature of the market is the dominance of Securitize as a tokenization platform. The majority of funds, including BCAP, ACRED, SPiCE VC, and multiple Hamilton Lane vehicles, rely on Securitize’s issuance and compliance infrastructure. As of 2025, Securitize manages the world’s largest tokenized U.S. Treasury product and holds over 70% of the tokenized securities market share. This platform dominance underscores the importance of regulatory compliance and institutional trust. At the broader level, the global RWA tokenization market (excluding stablecoins) has surpassed \$26 billion in 2025. Institutional funds account for a critical segment of this expansion.

## 7.2 Prevailing Structures and Regulatory Frameworks

The institutional fund sector overwhelmingly relies on structured offerings compliant with U.S. securities laws. Nearly every product in the dataset operates under Regulation D exemptions of the Securities Act, with regulatory oversight provided by the U.S. Securities and Exchange Commission. Funds are commonly domiciled in the United States, Singapore, the Cayman Islands, or the British Virgin Islands. Structurally, tokenization is achieved by integrating traditional fund vehicles with blockchain-based representations of ownership. This includes creating legal wrappers, appointing custodians, minting tokens on public blockchains such as Ethereum, Polygon, and Solana, and embedding compliance mechanisms. Securitize has become the leading provider of such infrastructure, working closely with regulators to legitimize tokenized offerings.

## 7.3 Economic Rationale & On-Chain Activity

The economic rationale behind institutional RWA funds is clear: they deliver access to yield, efficiency, and diversification. Following the 2022–2023 interest rate hikes, tokenized private credit and treasury products gained traction. Institutional funds build on this trend by providing blockchain-native exposure to strategies such as venture capital and private equity. From an efficiency standpoint, tokenization introduces near-instant settlement and transparent record-keeping. The ability to fractionalize holdings also democratizes access. Despite these strengths, on-chain activity tells a more cautious story. Most institutional RWA tokens exhibit low trading volumes and long holding periods. Liquidity is highly fragmented due to investor gating and compliance restrictions. In practice, this means tokenized funds today resemble long-term private placements more than liquid, tradable securities.

## 7.4 Synthesis: Strengths, Risks, and Outlook for Institutional Funds

Institutional tokenized funds present a unique mix of strengths and challenges. Their greatest strength lies in their regulatory credibility, as offerings are structured under Reg D exemptions and rely on platforms such as Securitize. They also appeal to yield-seeking investors by providing access to private credit and other income-oriented strategies. However, risks remain significant. Liquidity constraints are the most pressing, as most funds cannot be freely traded. Entry barriers are high, with minimum investments ranging from \$10,000 to \$100,000. Centralization risk persists as custodians and issuers act as critical gatekeepers. Looking ahead with a speculative note, we expect the outlook to be cautiously optimistic. Liquidity challenges may be addressed as hybrid markets emerge. Regulatory clarity is expected to improve as firms like Securitize continue to bridge the gap between DeFi and TradFi. With institutional capital increasingly comfortable engaging on-chain, tokenized institutional funds are poised to evolve from niche experiments into robust pillars of the global capital markets.

## 8 Asset Class Analysis: US Treasuries

Given the central role of US Treasuries in the global financial system, we separately study the market for tokenized US Treasuries. The tokenized US Treasuries market has grown into a \$7.5B segment across 47 products as of July 2025, driven by institutional adoption, Ethereum-based infrastructure, and integration into DeFi. The market is heavily concentrated, with the top three issuers—BlackRock, Franklin Templeton, and Ondo—holding nearly 70% of total market share. While the asset class offers benefits like fractional ownership and 24/7 liquidity, it also faces systemic risks from centralization and unclear regulation. Future expansion will depend on regulatory clarity, greater issuer diversification, and

improved interoperability.

## **8.1 Market Composition & Platform Dominance**

The tokenized US Treasuries market is highly concentrated, with a handful of major issuers controlling the majority of assets. BlackRock’s BUIDL fund dominates with approximately \$2.42B in market cap (35.76% share), followed by Franklin Templeton at \$780.5M (11.55%) and Ondo I at \$709.4M (10.5%). Combined, these top three issuers control nearly 70% of the market. This oligopolistic structure supports consistent liquidity and trust but also introduces systemic vulnerability. Ethereum has emerged as the overwhelmingly preferred blockchain, capturing roughly \$5B of the \$7.56B total market cap. This dominance stems from Ethereum’s extensive DeFi integration and established regulatory familiarity. Solana trails far behind with around \$300M in assets. While Ethereum’s lead reinforces network effects, it also concentrates technical and operational risk on a single chain.

## **8.2 Prevailing Structures and Regulatory Frameworks**

Tokenized Treasury products typically use special purpose vehicles (SPVs), LLCs, mutual funds, or limited partnerships as their legal wrappers. Jurisdictionally, the United States, British Virgin Islands (BVI), and Cayman Islands are the primary domiciles. US-based vehicles offer higher regulatory oversight and stronger investor protections. Offshore jurisdictions provide faster, more flexible formation processes but may come with trade-offs, such as weaker public disclosure requirements. The dominant pegging method is direct 1:1 asset custody of US Treasuries, often with daily or monthly reserve reporting. Some products include overcollateralization for added investor protection. Redemption processes vary, but many institutional-grade products only allow USD redemptions during market hours, whereas certain offerings support 24/7 redemptions in USDC via smart contracts. Audit

practices range from comprehensive, Big Four-led audits to basic attestations, leaving variability in transparency.

### **8.3 Economic Rationale & On-Chain Activity**

The tokenization of US Treasuries addresses several inefficiencies in traditional finance by enabling fractional ownership, round-the-clock market access, and seamless DeFi integration. These attributes make tokenized Treasuries attractive to both institutional and retail investors, though the market remains skewed toward institutional participation. Around 80% of the total market value is locked in products with high minimum investment thresholds. Retail-friendly products like BENJI and WTGXX exist but represent a smaller portion of the market. On-chain activity reflects the different investor profiles. Institutional holdings tend to be long-term, with low turnover, while retail participants often engage in higher-frequency transactions. Whale concentration is a notable trend, with some funds seeing single wallets controlling over 25% of the supply. Major mints and burns, often exceeding \$200M, have punctuated the market, typically in response to yield shifts or macroeconomic events.

### **8.4 Synthesis: Strengths, Risks, and Outlook for US Treasuries**

Tokenized US Treasuries offer a combination of stability, liquidity, and transparency. Backed by US government debt and often audited by top-tier firms, these products bring credibility to the DeFi ecosystem. Their integration as collateral in lending markets and their capacity for programmable yield distribution open up new use cases. The ability to fractionalize Treasuries and provide 24/7 access represents a significant innovation over traditional bond markets. Yet the asset class faces nontrivial risks. The heavy concentration of issuance among a few firms creates potential single points of failure. Jurisdictional and regulatory uncertainty adds further complexity. Going forward, growth will depend on expanding retail

access, achieving cross-chain interoperability, and clarifying global regulatory frameworks. If these challenges are addressed, tokenized Treasuries could evolve from a largely institutional niche into a mainstream fixed-income instrument.

## 9 Conclusion

As conclusion, we synthesize the cross-asset class findings of the preceding analysis to construct a holistic view of the tokenized real-world asset ecosystem. It consolidates the key themes, systemic risks, and structural patterns identified across the private credit, equities, global bonds, commodities, and institutional funds markets. Following this synthesis, a market-wide SWOT analysis is presented to systematically evaluate the ecosystem’s strategic position. The section concludes with a series of future predictions and identifies the most critical unanswered questions that will shape the next phase of the market’s evolution.

### 9.1 Summary of Key Cross-Asset Findings

A systematic analysis across the primary RWA asset classes reveals a market defined by three overarching themes: institutional consolidation, disconnection between issuance and liquidity, and the strategic use of regulatory arbitrage. First, the market is not a decentralized landscape but a series of concentrated oligopolies. In every major asset class, from commodities where two issuers control over 98% of the market to global bonds where a single issuer accounts for 84% of on-chain value, a handful of dominant, institutionally-focused platforms dictate market structure and standards. This consolidation of power suggests that trust, regulatory compliance, and established custodial relationships, hallmarks of traditional finance, remain the most critical factors for attracting significant capital, even within a DeFi context.

Second, a critical gap persists between the theoretical value proposition of tokenization

and its current on-chain reality. While the technology promises 24/7 global liquidity, our empirical analysis of on-chain data confirms this remains largely unrealized across most asset classes. The tokenized stocks sector provides the most salient example of this "liquidity paradox," where the vast majority of assets function as illiquid, static digital certificates with negligible secondary market activity. This pattern indicates that the market currently excels at primary issuance and redemption but has failed to bootstrap the robust secondary market infrastructure necessary to fulfill its core economic promise.

Third, the legal and structural architecture of the RWA ecosystem is fundamentally shaped by regulatory arbitrage. Issuers strategically select jurisdictions and legal wrappers—from Swiss DLT Act-compliant SPVs to U.S. statutory trusts under Regulation S—to optimize for speed-to-market and minimize compliance burdens. This has created a fragmented and complex legal landscape where the rights and protections afforded to token holders are inconsistent and often opaque. A pervasive failure to disclose third-party custodians across multiple asset classes further exacerbates this issue, introducing a significant layer of unquantifiable counterparty risk that directly contradicts the industry's narrative of enhanced transparency.

## 9.2 Market-Wide SWOT Analysis

An integrated assessment of the tokenized RWA ecosystem reveals a strategic position characterized by foundational strengths in access and efficiency, but undermined by significant weaknesses in market structure and transparency. The inherent strengths of the asset class lie in its proven ability to democratize access to previously siloed markets through fractionalization and to establish replicable legal blueprints for compliant, cross-border issuance. Furthermore, the dominant 1:1 asset-backing model provides a strong conceptual foundation of trust compared to more complex synthetic alternatives.

However, these strengths are counteracted by profound systemic weaknesses. The most

critical flaw is the near-total absence of organic, on-chain secondary market liquidity, which invalidates the core promise of 24/7 tradability. This is compounded by a high degree of centralization risk, where the operational or legal failure of a single dominant issuer or undisclosed custodian could trigger a cascading loss of confidence across an entire vertical. Moreover, the current models often strip token holders of fundamental rights, such as voting or direct dividend access, offering purely economic exposure and weakening the definition of digital ownership.

Despite these weaknesses, significant opportunities for future growth exist. The potential to integrate blue-chip RWAs as high-quality collateral within established DeFi lending protocols and structured product engines remains a largely untapped, multi-trillion-dollar opportunity. As regulatory clarity improves, particularly around tokenized ETFs, the market is well-positioned to attract substantial institutional capital seeking efficient, on-chain exposure. Conversely, the ecosystem faces considerable external threats. A targeted regulatory crackdown, particularly from the U.S. SEC on the prevailing interpretations of Regulation S, could cripple a significant portion of the market. Similarly, a major security exploit or insolvency event at a dominant, nontransparent issuer could prove to be a catastrophic, market-wide extinction-level event.

### **9.3 Future Research Directions**

We conclude by identifying several critical unanswered questions that future research may aim to address. First, what specific market structures and incentive mechanisms—be they automated market makers, partnerships with regulated digital asset exchanges, or novel liquidity mining programs—are required to successfully bootstrap genuine secondary market liquidity for these on-chain assets? Second, can a sustainable model emerge that provides the benefits of on-chain composability without forcing investors to sacrifice fundamental asset rights, such as governance and direct legal claims? Third, will the market pivot from

nascent, single-name tokenized assets toward more diversified products, such as tokenized ETFs and regulated money market funds, with the promise of offering better diversification and risk factor exposure or management?<sup>5</sup> Finally, as global regulators continue to develop frameworks for digital assets, will the market consolidate around a single dominant legal standard, or will it remain a fragmented ecosystem of competing jurisdictional models? The answers to these questions will ultimately determine whether the tokenization of real-world assets evolves from a niche financial experiment into a foundational pillar of the global economic infrastructure.

## Project Team

Under the supervision of Jiasun Li, this report is made available by the joint efforts of the project team members, including Aniketh Malipeddi, Adhiraj Chhoda, Ajay Vinjamuri, Alvin Wang, Anna Xiao, Arjun Krishnamurthy, Avanthika Thodupunuri, Avishi Anurag, Calvin R. Hurwitz, Colin Adam Kim, Ethan Cui, Eythan Huang, Isha Joshi, Kevin Lu, Luvyakrish Manche, Naveen Balakrishnan, Nihar Anand, Pranav Mathur, Rahul Dham, Sahas Manchireddy, Sahil Aggarwal, Sathvik Vaka, Smaran Bonthula, Srivatsa Veera, Ved Patil, Vikhyat Duddilla, Zoeb Izzi, and Ishan Rekhi.

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<sup>5</sup>See for example [Li and Yi \(2019\)](#); [Liu, Tsyvinski and Wu \(2022\)](#) on the factor structures in cryptoasset returns.

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